

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURJADA
VIZIANAGARAM
B.Tech III Year I Semester - Computer Science and Engineering
DESIGN AND ANALYSIS OF ALGORITHMS (R23)
Unit-wise 2 Marks and 10 Marks Important Questions of 1 to 3 units**

UNIT - I: Introduction

2 Marks Questions:

- Define Algorithm. What are the criteria that an algorithm must satisfy?
- Explain pseudo-code conventions used for writing algorithms.
- Define asymptotic notations: Big-Oh (O), Omega (Ω), and Theta (Θ).
- What is Amortized Complexity? Give an example.
- Distinguish between Space Complexity and Time Complexity.
- Explain the concept of Performance Analysis and Practical Complexities' Performance Measurement.

10 Marks Questions:

- Explain in detail about asymptotic notations with mathematical definitions, graphical representations, and examples.
- Discuss the general framework for analyzing the time and space complexity of recursive and non-recursive algorithms with suitable examples.
- Explain pseudo-code conventions in detail. Write an algorithm for finding the maximum and minimum elements in an array and analyze its time complexity.
- Write short notes on: (a) Amortized Complexity (b) Space Complexity (c) Practical performance measurement.

UNIT - II: Decrease-and-Conquer, Divide-and-Conquer, and Transform-and-Conquer

2 Marks Questions:

- Define Decrease-and-Conquer technique. Give examples.
- State the general principle of Divide-and-Conquer.
- What is Strassen's Matrix Multiplication? What is its time complexity?
- Define heap and heap sort.
- What is pre-sorting? How does it help in solving searching problems?
- Explain the working principle of Merge Sort.

10 Marks Questions:

- Explain Merge Sort algorithm using Divide-and-Conquer strategy. Trace it for the array [38, 27, 43, 3, 9, 82, 10] and derive its worst-case time complexity.
- Explain Quick Sort algorithm. Discuss its partitioning strategy and analyze its best, average, and worst-case time complexities.
- Explain Strassen's Matrix Multiplication in detail. How does it improve upon standard matrix multiplication complexity?
- Discuss Decrease-by-a-Constant-Factor and Variable-Size-Decrease algorithms with suitable sorting examples (e.g., Insertion Sort).

UNIT - III: The Greedy Method**2 Marks Questions:**

- What is the Greedy Method? State its core characteristics.
- Define Knapsack Problem. How does Greedy approach solve it?
- State Job Sequencing with Deadlines problem.
- What is a Minimum Cost Spanning Tree (MCST)?
- Differentiate between Prim's and Kruskal's algorithms.
- What are Optimal Merge Patterns?

10 Marks Questions:

- Explain the general method of Greedy approach. Solve the continuous knapsack problem for $n = 3$, $M = 20$, $(p_1, p_2, p_3) = (25, 24, 15)$, and $(w_1, w_2, w_3) = (18, 15, 10)$.
- Explain Kruskal's algorithm for finding Minimum Cost Spanning Tree with an example graph and analyze its time complexity.
- Explain Prim's algorithm for finding Minimum Cost Spanning Tree with an illustrative example.
- Discuss Dijkstra's algorithm for Single Source Shortest Paths with a suitable example graph.